

LECTURE 02

The OSI Reference Model

From the TCP/IP model to a more detailed view of networking

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PART ONE

Where We Left Off

Before anything new, we return to the model from Lecture 01 and follow one message all over again.

LAST TIME

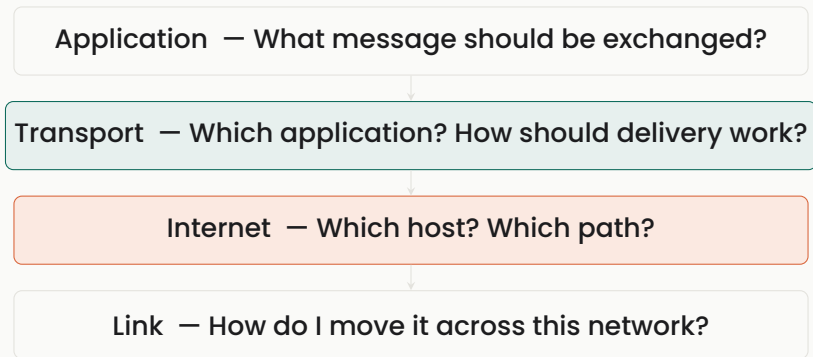
We split networking into four layers

Networking is one enormous problem. In Lecture 01 we made it manageable by cutting it into four layers, each with its own single job:



Application → Transport → Internet → Link

Each layer answered one question



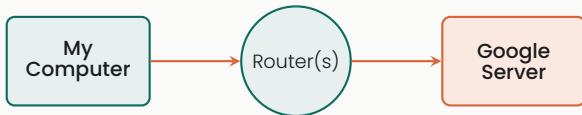
Keep these four questions in your head. Everything today is built on them.4 / 41

THE SAME STORY

“I type google.com and press Enter”

We will follow the exact same journey as last time, just drawn a little more concretely.

My browser wants a page from Google. That request does not jump straight across — it passes through one or more routers on the way.



My Computer → Router(s) → Google Server

Provides network services to applications

Its responsibilities:

- Defines application-level communication
- Structures the data being exchanged
- Uses protocols such as HTTP, DNS, SMTP

Delivers data from one application to another

- **Application identification**
ports and application endpoints
- **Segmentation**
break and reassemble data
- **Reliability**
detect loss and recover when supported
- **Flow control**
protect the receiver
- **Congestion control**
adapt to network conditions
- **Multiplexing**
carry many conversations together
- **Demultiplexing**
deliver each conversation correctly

Delivers packets from one host to another

Its responsibilities:

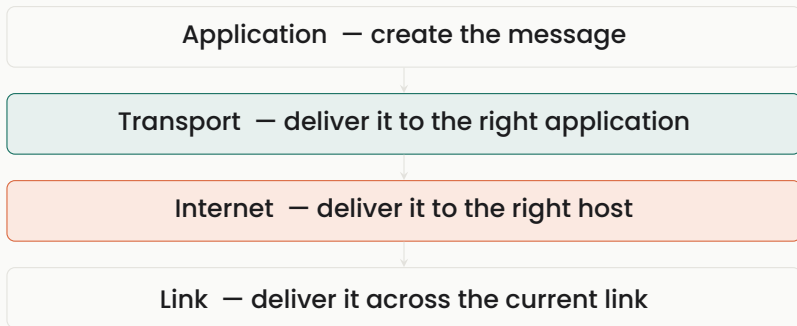
- Provides IP addressing
- Determines how packets are routed
- Forwards packets across networks

Moves data across a single link

Its responsibilities:

- Uses local addressing such as MAC addresses
- Creates frames
- Controls access to the medium
- Detects transmission errors

One message, four jobs



This is the mental model. Hold on to it.

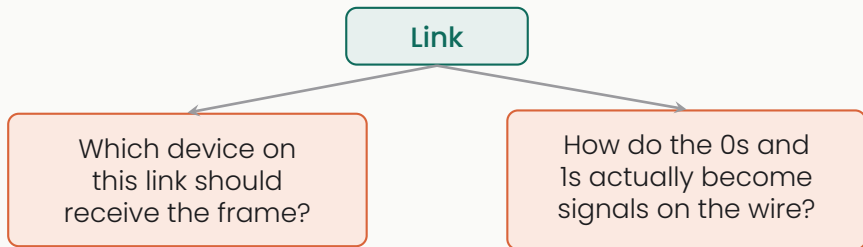
A NEW QUESTION

**We divided networking into four layers.
But is four the only way to divide it?**

Four layers was a choice, not a law of nature. What if we wanted to separate some responsibilities more carefully?

The Link layer hides two jobs

The single TCP/IP Link layer moves data across one hop. Look closer — could it really be two different jobs bundled together?



Both jobs live on the very same link, yet answer different questions.

THE IDEA

Introducing the OSI Reference Model

The OSI Reference Model asks a simple question:

“Can we break the communication problem into seven more specific responsibilities?”

OSI stands for Open Systems Interconnection. It is not a program you install — it is a way of organising thinking about a network.

A model tells us

what responsibilities exist, and how we organise them.

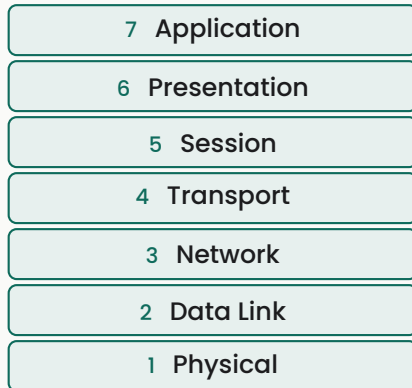
We will make the difference between a model and a real protocol very clear later.

SLOWLY NOW

The seven OSI layers

Read them from the top (closest to the user) down to the bottom (closest to the wire).

Do not memorise them yet. First we only want to know what each one is responsible for.



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PART TWO

Understanding the New Boundaries

What is each of the seven layers actually responsible for — and which ideas are genuinely new?

ONE QUESTION PER LAYER

Seven layers, seven questions

Application

What network service does the application need?

Presentation

What does the data mean / look like?

Session

How is the communication session managed?

Transport

Which application, and how should delivery work?

Network

Which host, and how does it travel between networks?

Data Link

How does it cross the current local link?

Physical

How are the actual bits / signals transmitted?

Notice: some of these questions we already know. A few are new.

OSI breaks “Application” into three

The one TCP/IP Application layer becomes three OSI layers, each answering a narrower question:

Application

What service does
the app need?

Presentation

How is the data
represented?

Session

How is the conversation
managed?

Let us meet the two new ones: Presentation and Session.

“Do both sides agree what the bytes mean?”

Even if two computers receive the same bytes, they still need to agree on what those bytes mean:

- Data representation and format
- Encoding and translation
- Compression
- Encryption and decryption

Two people can exchange the same facts — but must first agree on a common language.

Quick check If two computers exchange the same information but use different data representations, which OSI layer helps us think about that problem?

Presentation.

“How is an ongoing conversation managed?”

Sometimes communication is not one isolated message but an ongoing conversation. The Session layer is about managing that conversation:

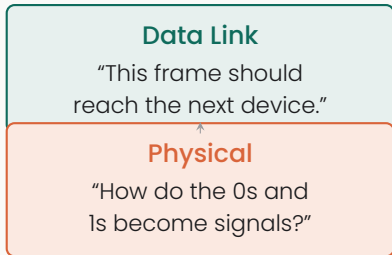
- Establishing a session
- Maintaining it
- Coordinating the exchange
- Ending it cleanly

Important warning

In the real Internet there is usually no single neat “Session protocol.” These jobs are spread across several protocols and components. The layer names a responsibility, not one program.

Below the frame: the actual signals

Last time, the Link layer stopped at “get this frame to the next device.” But how does a frame physically travel through a cable or through the air? OSI gives that its own layer.



The Physical layer covers:

- Transmitting raw bits
- Electrical, optical, radio signals
- The physical medium
- Connectors and interfaces

Quick check Which OSI layer is responsible for the actual transmission of signals?

Physical.

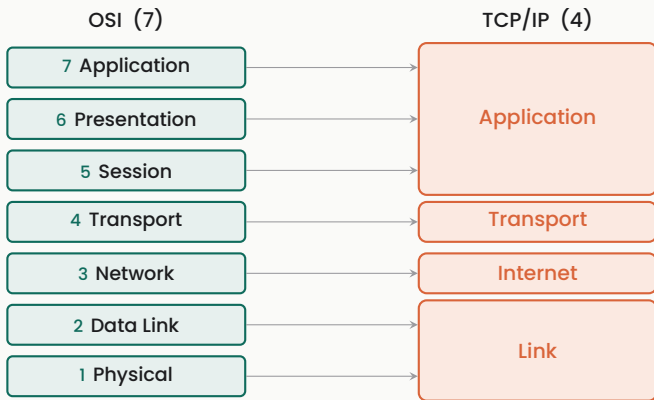
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PART THREE

Two Models, One Journey

The same message, seen through two different arrangements of the same responsibilities.

How OSI lines up with TCP/IP



The layers line up — but not perfectly

- TCP/IP **Application** covers OSI *Application + Presentation + Session*
- TCP/IP **Transport** ↔ OSI *Transport*
- TCP/IP **Internet** ↔ OSI *Network*
- TCP/IP **Link** covers OSI *Data Link + Physical*

The key idea

The network did not change. We are looking at the same communication problem through a different arrangement of responsibilities. The boundaries differ; the problem does not.

One communication — two models

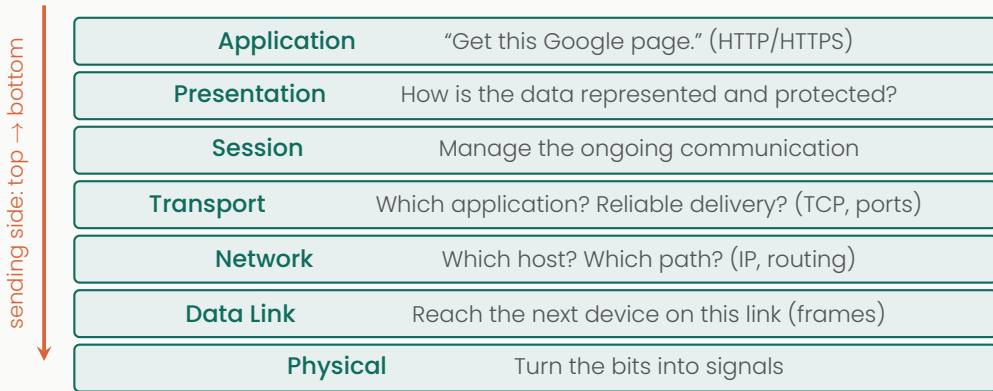


The communication is the same. The general responsibilities are the same. Only the boundaries between them are drawn differently.

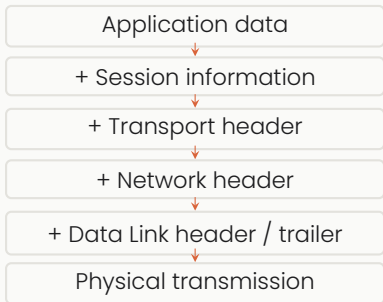
Quick check Are OSI and TCP/IP two different Internets?

No — one Internet, two ways of describing it.

Down through all seven OSI layers



Each layer adds what it needs — conceptually

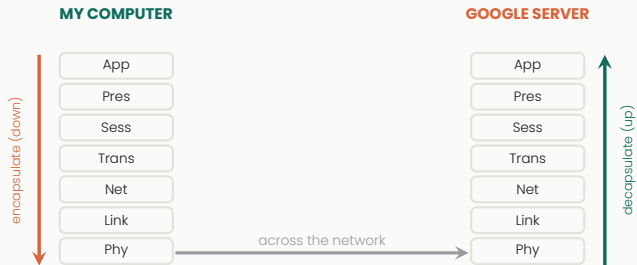


Read this carefully

This is a conceptual picture of layer responsibilities. It does not mean every OSI layer adds a real, separate header in every real protocol stack. Some of these are ideas, not literal bytes on the wire.

BOTH ENDS

Down on the way out, up on the way in



The sender wraps layer by layer going down; the receiver unwraps layer by layer going up.

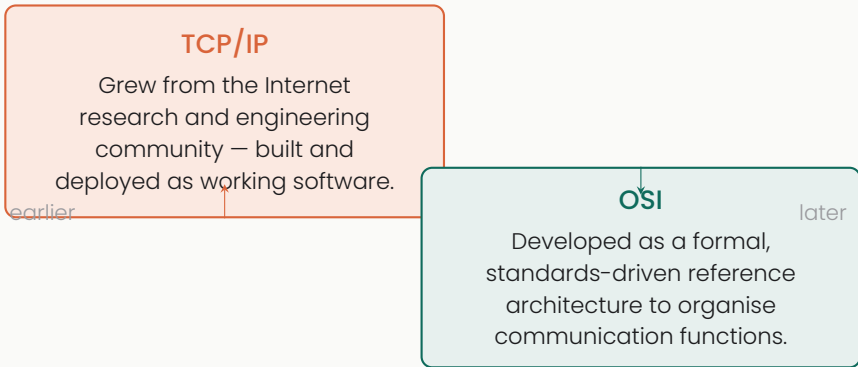
THE MYSTERY

If TCP/IP already works, why do we even have OSI?

The answer is not “old versus new.” The two came from different places, for different reasons.

WHERE THEY CAME FROM

Two different origins



Different contexts, different goals – not simply one replacing the other. 29 / 41

Three different things, often confused

OSI model

A conceptual /
reference architecture.

OSI protocols

Actual protocols
designed around
that architecture.

TCP/IP

A protocol suite
already being built
for the Internet.

A model says how to organise. A protocol says the exact rules
devices follow.

A combination of factors — not one slogan

It is tempting to say “OSI was just too complicated.” The honest picture is a mix of practical factors:

- Working TCP/IP was *already deployed*
- OSI protocols arrived relatively late
- Real Internet growth had momentum
- Implementation and deployment were hard
- Complexity in places
- A large ecosystem already spoke TCP/IP

Timing, deployment, implementation, and ecosystem — together, not any single cause.

SO WHAT HAPPENED?

The protocols faded — the model did not

OSI protocols

did not become the Internet's dominant protocol stack.

vs

OSI model

remains widely used as a way to reason about networks.

“Did not win as protocols” $\neq$ “useless.”

A shared language for networking

- Finer separation of responsibilities
- A common vocabulary everyone shares
- Helps you troubleshoot networks
- Helps compare technologies
- Helps decide where a problem belongs
- Separates ideas even when real protocols cross the lines

When an engineer says “that’s a layer 3 problem,” this is the model they mean.

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PART FOUR

Using the Model

Layering is not just for exams. It is how we find where a problem lives.

“The website isn’t loading. Where’s the problem?”

check from the bottom up ↑

Physical	Is the cable plugged in? Is Wi-Fi actually connected?
Data Link	Can the device talk on the local network at all?
Network	Can we reach the destination IP address?
Transport	Can we open the connection we need?
Application	Is HTTP / HTTPS itself responding?

Layers let us check one thing at a time instead of guessing.

Model versus protocol

A model

tells us what responsibilities exist and how we organise them.

A protocol

tells us the exact rules devices follow to communicate.

OSI

TCP/IP

HTTP

TCP

IP

Ethernet /
Wi-Fi

reference model

protocol suite + layers

application protocol

transport protocol

network / internet protocol

link technologies

TCP/IP and OSI, at a glance

Question	TCP/IP	OSI
Number of layers	4	7
Main purpose	Practical suite	Reference model
Application-related layers	1	3
Transport	Transport	Transport
Network layer	Internet	Network
Local transmission	Link	Data Link + Physical
Dominant on the Internet?	Yes	No
Useful for learning?	Yes	Yes

THE TAKEAWAY

Seven questions, not seven words to memorise

Application	What are we trying to communicate?
Presentation	How is the information represented?
Session	How is the conversation managed?
Transport	Which application, and how should delivery work?
Network	Which host, and which path?
Data Link	How do I cross this local link?
Physical	How do the bits become signals?

Understand the reason for each boundary, and the names take care of themselves.

YOUR TURN

Think

You open a video call and the picture freezes, but the chat text still sends fine.

- Which layers would you not suspect first, and why?
- Does thinking in layers make the search easier?

There is no single “right” answer — we want the reasoning.

THE ONE IDEA TO KEEP

A network is a big problem; layering breaks it up

Application	What are we trying to communicate?
Presentation	How is the information represented?
Session	How is the conversation managed?
Transport	Which application, and how should delivery work?
Network	Which host, and which path?
Data Link	How do I cross this local link?
Physical	How do the bits become signals?

TCP/IP and OSI are two ways of organising the same communication problem.

WHERE WE ARE NOW

Lecture 01: we needed layers. Lecture 02: layers can be drawn in more than one way.

The layers are a tool for humans to reason about networking. Real protocols do not always respect the boundaries perfectly — and that is fine.